

Exercise 01 – Python

PART 01

1. Complete below Task using Python programming,
 - I. Add 20 and 15 and print the result.
 - II. Subtract 12 from 50 and print the result.
 - III. Multiply 8 and 7 and print the result.
 - IV. Divide 100 by 25 and print the result.
 - V. Find the remainder when 17 is divided by 5 and print the result.
2. Write Python Program to check **Validity of Triangle**. Ask the user for **three sides of a triangle**, check if they form a valid triangle, and print the result. (Hint: - Triangle Inequality Theorem: The sum of the lengths of any two sides must be greater than the length of the third side.)
3. Create a program to simulate simple calculator:
 - Asks user for two numbers
 - Asks user for operation (+, -, ×, ÷)
 - Print the result with appropriate message.
4. Write a Python program using a **for loop** to enter marks for **5 subjects** and calculate the **total marks**.
5. Write a Python program that continuously asks the user to enter a password using a **while loop**. The program should keep requesting the password until the correct password "**python123**" is entered. Once the correct password is entered, display the message "**Access granted**".

PART 02

1. **Complete below Task using Python programming,**
 - I. Compare two numbers ($x = 15$, $y = 10$) and
 - (a) Check if x is greater than y .
 - (b) Check if x is equal to y and print the result.
 - (c) Check if x is not equal to y and print the result.
 - II. Compare 10 and 20 using $\leq$ and print the result.
 - III. Check if both $5 > 3$ and $10 > 5$ are true and print the result.

2. **Create a program to simulate BMI calculator:**

- Asks user for weight (kg) and height (m)
- Calculate BMI using formula: $\text{weight} / (\text{height} \times \text{height})$
- Classifies BMI (Underweight: <18 , Normal: $18-25$, Overweight: >25)
- Print BMI value and classification

3. **Create Student Grade Management System:**

- Ask for student's name.
- Take marks for 3 subjects.
- Calculate the total mark out of 100 (percentage) and print detailed mark sheet with all information.

Determine grade as below.

- A: $\geq 75\%$
- B: $60-75\%$
- C: $45-60\%$ - D: $30-45\%$
- F: below 30%

4. Write a Python program that takes a number **n** as input from the user and uses a **for loop** to calculate the **sum of the first n natural numbers**. Finally, display the result.
5. Write a Python program for an **online shopping checkout system** using **while loop**.
 - The program should keep asking the user to enter the **price of items**.
 - The checkout process should continue until the **total bill amount reaches Rs.1000 or more**.
 - Once the total is reached, display "**Checkout complete**" and print the **total amount to be paid**.